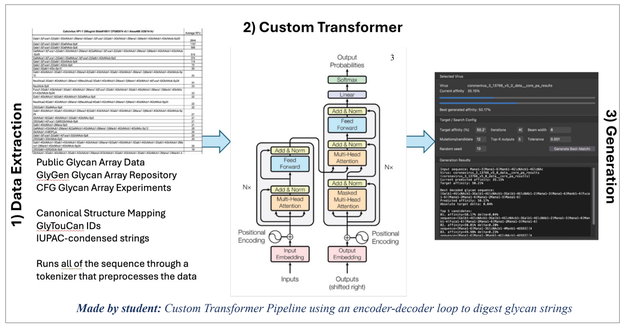
GLYDE

A Novel Domain-Aware, Topology-Biased Glycan Language Model for Viral Receptor Binding

Credit Line: All images/graphs/pictures are created by the student unless stated otherwise

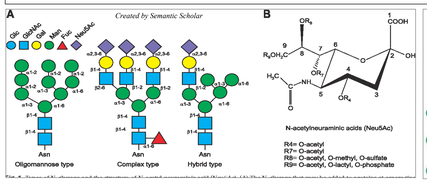
VISUAL SUMMARY



Made by student: *Custom Transformer Pipeline using an encoder-decoder loop to digest glycan strings*

INTRODUCTION

- Viruses identify host cells by reading surface glycans, where binding is primarily driven by **terminal motifs and linkage chemistry** rather than the full glycan structure (classic examples include influenza's preference for **α2-3 vs α2-6 sialylation**). While genomic/proteomic language models like **Evo2** have shown how powerful large-scale sequence learning can be, existing glycan LMs largely treat glycans as generic sequences or graphs and rely purely on data alone to learn these patterns.
- Introducing **Glyde,** a **topology-biased glycan language model** that encodes biological knowledge directly into the transformer architecture, emphasizing terminal residues, linkage types, and branch depth through attention bias and structure-aware pooling. Using curated publicly available glycan array datasets and canonical glycan structures, we evaluate Glyde on viral binding **prediction, classification, and retrieval**, testing structural holdout splits to measure true generalization beyond simple motif memorization.



OBJECTIVES

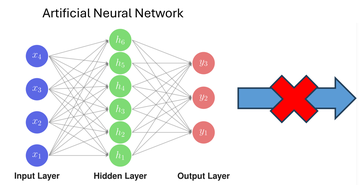
Problem: Predicting virus–glycan binding is difficult because glycans are highly branched, non-linear structures where binding is governed by terminal motifs and linkage chemistry rather than full sequence composition.

Question: How can a topology-aware deep learning framework improve prediction and generation of glycan structures for viral binding beyond sequence-based models?

Claim: Glyde, a topology-biased glycan language model, improves performance by explicitly encoding terminal residues, linkage types, and branch depth into transformer attention and pooling mechanisms.

Objectives / Brainstorming & Solution:
1) Represent glycans using both **token sequences and topology feature vectors** (depth, linkage, terminal distance).
2) Integrate **topology-aware attention biases** to prioritize biologically relevant interactions.
3) Implement **structure-aware pooling** to emphasize terminal residues critical for binding.
4) Train on **glycan array datasets** with structural holdout splits to evaluate true generalization.
5) Validate performance on **binding prediction, classification, and retrieval tasks**.

PROBLEM FRAMING



Predicted Binding Probabilities

#	Strain	Score
1	A/Pennsylvania/08/2008 (H1N1)	0.8809
2	A/South Carolina/1/1918 (H1N1)	0.8050
3	H3N8 WT (10000 HAU/ml)	0.7721
4	H6N6 WT (3200 HAU/ml)	0.7689
5	A/BCM/2/1969	0.6586

Made by Student: *Artificial Neural Network Architectures cannot effectively learn relations in glycan structures as well as transformers*

Traditional neural network architectures treat glycans as **flat inputs**, limiting their ability to capture the branching structure, linkage patterns, and terminal motifs that govern viral binding. As a result, they struggle to model long-range and non-linear dependencies within glycan structures. Transformer-based models, through self-attention, provide a more flexible framework for learning these complex structural relationships.

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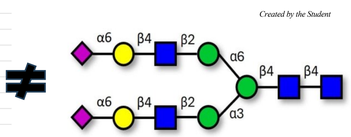
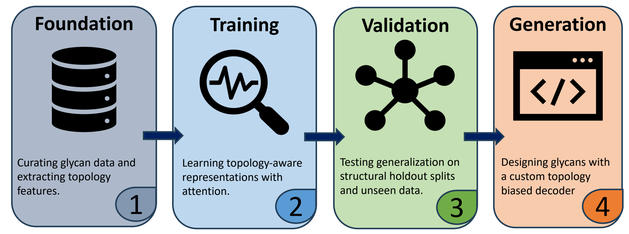
5 (3OSO3)(6OSO3)Galb1-4GlcNAcb-Sp0
6 (3OSO3)GalNAcb1-4(3OSO3)GlcNAc-Sp8
7 (3OSO3)GalNAcb1-4(Fuca1-3)GlcNAcb-Sp8
8 (3OSO3)GalNAcb1-4GlcNAc-Sp8
9 (3OSO3)Galb-Sp8
10 (3OSO3)Galb1-3(Fuca1-4)GlcNAca-Sp0
11 (3OSO3)Galb1-3(Fuca1-4)GlcNAcb-Sp8
12 (3OSO3)Galb1-3GalNAca-Sp8



Sequence representations flatten glycans into linear tokens, losing critical structural information such as branching patterns, linkage types, and terminal residues. However, viral binding is determined by these structural features, not just the order of components. As a result, **sequence-based models cannot fully capture the interactions** that govern glycan–virus binding.

COMPUTATIONAL PIPELINE: ENGINEERED FRAMEWORK



Data Visualization: Viral Receptor Binding (1)

primscreen_PA_v1_263_12072005	James Paulson	Influenza virus A/New_York/1/18 (H1N1)	Influenza virus A/New_York/1/18 (H1N1)
primscreen_PA_v1_262_12072005	James Paulson	Influenza virus A/South Carolina/1/1918 (H1N1)	Influenza virus A/South Carolina/1/1918 (H1N1)
primscreen_PA_v1_261_12072005	James Paulson	Influenza virus A/Duck/Ukraine/1/1963 (H3N8)	Influenza virus A/Duck/Ukraine/1/1963 (H3N8)
primscreen_PA_v1_260_12072005	James Paulson	Influenza virus A/Moscow/10/1999 (H3N2)	Influenza virus A/Moscow/10/1999 (H3N2)
primscreen_PA_v1_259_12072005	James Paulson	Influenza virus A/Texas/36/1991(H1N1)	Influenza virus A/Texas/36/1991(H1N1)
primscreen_PA_v1_325_12072005	James Paulson	Influenza virus A/Duck/Alberta/35/1976 (H1N1)	Influenza virus A/Duck/Alberta/35/1976 (H1N1)
primscreen_PA_v1_320_12072005	James Paulson	Influenza virus A/Vietnam/1203/2004 (H5N1)	Influenza virus A/Vietnam/1203/2004 (H5N1)
primscreen_PA_v2_187_11072005	Gillian M. Air	A/Wyoming/3/2003 (Influenza virus)	Virus
primscreen_PA_v2_239_11022005	Gillian M. Air	A/California-like Influenza H3N2	1391, 1992
primscreen_PA_v2_238_12062005	Gillian M. Air	A/California-like Influenza H3N2	1391, 1992
primscreen_PA_v2_188_12062005	Gillian M. Air	A/Wyoming/3/2003 (Influenza virus)	Virus
primscreen_PA_v2_186_12082005	Gillian M. Air	A/Philippines/82 (Influenza virus)	Virus
primscreen_PA_v2_257_10252005	Gillian M. Air	Influenza virus A/Oklahoma/323/03 (H3N2) - B	Influenza virus A/Oklahoma/323/03 (H3N2) - B
primscreen_PA_v2_168_10242005	Gillian M. Air	Influenza virus A/Oklahoma/323/03 (H3N2) - B	Influenza virus A/Oklahoma/323/03 (H3N2) - B
primscreen_PA_v2_167_10242005	Gillian M. Air	Influenza virus A/Oklahoma/323/03 (H3N2) - A	Influenza virus A/Oklahoma/323/03 (H3N2) - A
primscreen_PA_v2_169_10242005	Gillian M. Air	Influenza virus A/Oklahoma/323/03 (H3N2) - C	Influenza virus A/Oklahoma/323/03 (H3N2) - C
primscreen_PA_v2_321_12072005	James Paulson	Influenza virus A/Duck/Singapore/Q-F119-3/19 (H	Influenza virus A/Duck/Singapore/Q-F119-3/19 (H
primscreen_PA_v2_310_12072005	James Paulson	Influenza virus A/Vietnam/1203/2004 (H5N1)	Influenza virus A/Vietnam/1203/2004 (H5N1)
primscreen_918	Gillian M. Air		Influenza Virus A/Oklahoma/370/05, 4 degree, T

Training and Validation System Information

Languages: Python, LaTeX
Libraries: NumPy, PyTorch, Tkinter, scikit-learn, Matplotlib, Pandas
Datasets:
- GlyGen Glycan Array Repository: Curated glycan structures and glycan interaction data
- CFG Glycan Array Experiments: Experimental glycan microarray data measuring viral protein binding (e.g., influenza hemagglutinin)
- NCBI Glycans: Used to view the SNFG structure of glycans to make detailed illustrations

Inference Specifications:

Personal Machine	NVIDIA DGX-1
M3 Pro Chip, 18 GB RAM	8 V100s SXM4, 8 TB RAM

DGX-1 used for **training the model**, while the Personal Machine was used for **inference tasks in the GUI**

Viral Binding Prediction Outputs (2)

```
01. H1N1 Influenza
A/Pennsylvania/08/2008- 6.2e5 pfu/ml
0.9083 (90.83%)
02. Influenza A/South Carolina/1/1918
(H1N1)  0.8624 (86.24%)
03. A/OK/447/08  0.7599 (75.99%)
04. EAvian Influenza
A/chicken/Pennsylvania/13609/1993 Low
pathogenic H5N2  0.7562 (75.62%)
05. Influenza A/Pennsylvania/08/2008
0.7544 (75.44%)
06. H1N1 Influenza
A/Pennsylvania/08/2008- 6.2e6 pfu/ml
0.7351 (73.51%)
07. H3N8 WT-10000HAU/ml  0.7333
```

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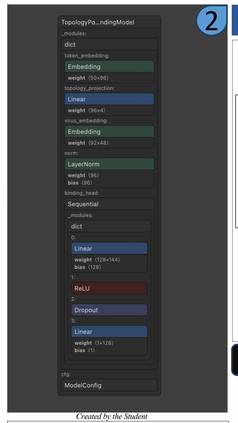
The model outputs predicted binding probabilities between glycans and viral strains, ranking candidates by affinity. **Higher scores indicate stronger predicted interactions**, with top-ranked glycans reflecting key structural features such as terminal residues and linkage patterns. This enables identification of biologically relevant binding motifs and prioritization of candidates for further analysis.

(2)



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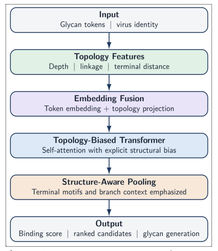
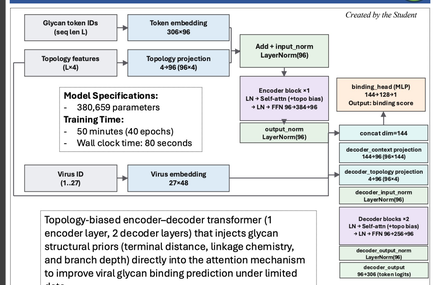
- ✓ **3 Transformer Layers** (Encoder + 2 Decoders); Each block has self-attention + feed forward networks



- ✓ Traditional methods use **mean/max pooling**, which treat all positions equally or select only the strongest signal
- ✓ Glyde uses **structure-aware pooling** aggregates token representations into a global glycan embedding
- ✓ Uses **weighted pooling** to emphasize terminal residues and key structural positions
- ✓ Captures **branch-level context** rather than uniform averaging across tokens
- ✓ Produces a **compact representation** for binding prediction and ranking

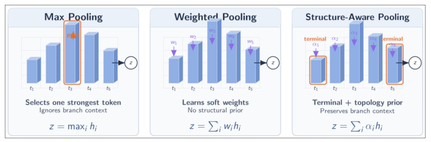
Custom Transformer Architecture (3)



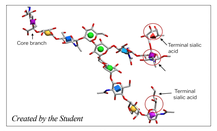
Topology-biased encoder–decoder transformer (1 encoder layer, 2 decoder layers) that injects glycan structural priors (terminal distance, linkage chemistry, and branch depth) directly into the attention mechanism to improve viral glycan binding prediction under limited data.

Topology Biased Attention

Component	Mathematical Formulation
Topology Feature Vector	$f_i = \begin{bmatrix} d_i \\ t_i \\ l_i \\ r_i \end{bmatrix}$
Topology-Aware Token Representation	$x_i = \mathrm{LayerNorm}(\mathrm{Emb}(t_i) + W_{\mathrm{topo}} f_i)$
Pairwise Topology Interaction	$g_{ij} = \begin{bmatrix} 1 - \lvert d_i - d_j \rvert \\ t_i t_j \\ l_i l_j \\ 1 - \lvert r_i - r_j \rvert \end{bmatrix}$
Topology-Biased Attention Score	$\mathrm{score}_{ij}^{(h)} = \frac{q_i^{(h)} k_j^{(h)}}{\sqrt{d_h}} + \lambda \left(w_h^\top g_{ij} \right)$
Topology-Biased Pooling Weight	$\alpha_i \propto 1 + \frac{w_d}{1+d_i} + w_t t_i + w_l l_i + \frac{w_r}{1+r_i}$



These topology features influence token embeddings, attention interactions, and pooling weights, allowing the model to prioritize the structurally important glycan residues such as motifs that play a role in viral binding.



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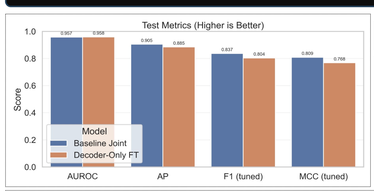
COMPUTATIONAL PIPELINE: STATISTICAL ANALYSIS (3)

Classification Metrics



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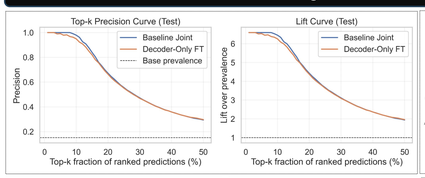
$$\mathrm{AUROC} = \Pr\left(f_\theta(g^+, v^+) > f_\theta(g^-, v^-)\right)$$

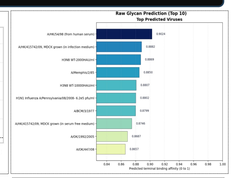
AUROC measures how well the model ranks true glycan-virus binding interactions above non-binding interactions. For any pair consisting of a real binding pair and a non-binding pair, **AUROC is the probability that the model will assign a higher score to the true binding interaction.** Compared with the baseline joint, decoder-only fine-tuning typically has better task-specific generation but has worse generalization performance.

Model Variant	Val AUROC	Test AUROC	Test AP	Test F1
Baseline (token + virus embedding + MLP)	0.90	0.88	0.62	0.64
+ Topology features (depth / linkage / terminal distance)	0.93	0.91	0.72	0.70
+ Topology-biased attention	0.96	0.94	0.83	0.77
+ Joint training (best binding checkpoint)	**0.975**	**0.957**	**0.905**	**0.837**

- ✓ **AUROC = 1.0** → perfect ranking of binding vs non-binding pairs
- ✓ **AUROC = 0.96 (my model)** → the model correctly ranks binding interactions above non-binding ones about **96% of the time**.

Ranking Metrics





Top-k Precision: Shows how accurate the model is among the highest-ranked predictions. Precision is **near 1.0 for the top predictions**, meaning the model correctly identifies true glycan–virus binding pairs near the top of the ranking.

Lift Curve: Measures improvement over random selections. The model achieves **~6× enrichment at the highest ranks**, meaning the top predictions contains far more true binders than expected by chance. This indicates the model is good at prioritizing relevant interactions.

Top-ranked predictions align with known glycan–virus binding patterns, showing that the model sees key structural features such as terminal residues and linkage chemistry. This means the **model learns biologically meaningful signals** and reliably prioritizes true binding interactions among its highest-confidence predictions.

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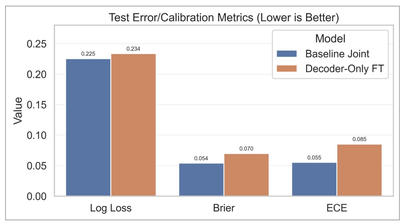
Calibration Metrics

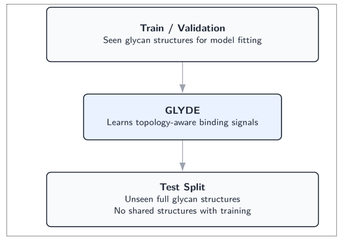


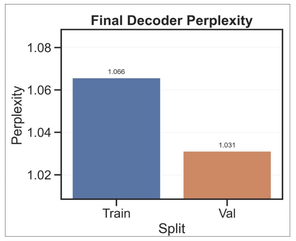
- ✓ **Lower log loss, Brier score, and ECE** indicate strong calibration, meaning predicted probabilities closely match true outcomes and reflect reliable confidence estimates.
- ✓ **Log loss:** penalizes overconfident errors → lower is better.
- ✓ **Brier score:** measures probability accuracy → lower is better.
- ✓ **ECE:** measures confidence vs. reality alignment → lower is better.

Overall, these metrics demonstrate that the model produces reliable and well-calibrated confidence estimates, making predictions more trustworthy for downstream biological analysis and decision-making.

All Images/Graphics/Figures made by Students on this section

Generalization / Robustness





Structural holdout evaluation ensures that **full glycan structures are not shared between training and test sets**, forcing the model to generalize beyond memorized motifs.

Decoder perplexity measures how well the **model predicts the next token in a sequence**, with lower values indicating more confident and accurate predictions.

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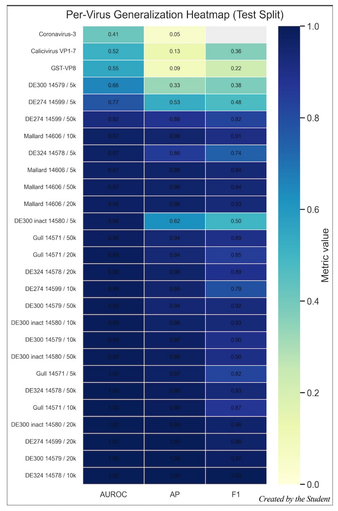
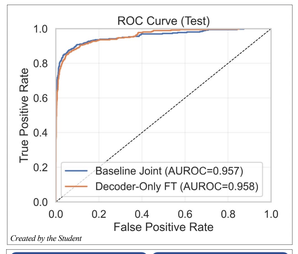
RESULTS AND BENCHMARKING (4)





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Encoder / MLP	Transformer
Token emb: 96 Virus emb: 48 Hidden: 128 Dropout: 0.1	Layers: 1 Heads: 4 FFN mult: x4 Topo attn: 0.75
Training	**Topology Bias**
Batch: 256 LR: 0.001 Weight decay: 0.0001 Epochs: 40 | Seed: 13	Depth: 0.35 Terminal: 0.5 Linkage: 0.2 Distance: 0.25
Decoder	**Input / Split**
Hidden: 256 Layers: 2 Loss wt: 0.5 Label smooth: 0	Max tokens: 32 Unknown token: [UNK] Split: random Tags: train / val / test

- ✓ **0.957 Test AUROC, 0.905 AP**, demonstrating strong and reliable predictive performance across evaluation metrics
- ✓ Consistent performance gains from **topology-aware features** compared to sequence-based baseline models
- ✓ Accurate ranking of **high-affinity glycans**, effectively prioritizing biologically relevant candidates
- ✓ Robust performance under **structural holdout splits**, indicating true generalization beyond motif memorization
- ✓ Well-calibrated predictions with **low Brier score and ECE**, ensuring reliable and confident outputs

CONCLUSIONS

- Overall, the results from Glyde support the hypothesis/claim of my project: incorporating biological structure directly into a glycan language model improves viral receptor-binding prediction. By **encoding topology-aware features such as terminal residues, linkage chemistry, branch depth, and terminal distance**, the model outperforms simpler baselines very consistently, with the final and best model reaching about **0.957** test AUROC and **0.905** test AP.
- At the same time, the training shows that careful regularization and stronger evaluation are still very important, helping to create **more structural holdout splits, virus-holdout testing, and ablation studies**. All in all, these findings show that small, domain-aware glycan models like Glyde can **capture biologically meaningful binding patterns in a way that is very practical** for research labs and institutions.

FUTURE WORK

- By combining biological knowledge with an efficient transformer architecture, Glyde allows **models to pick up patterns and have accurate predictions of glycan–virus interactions** that can guide experimental design processes. In labs, this type of model could help them prioritize which glycans to test, identify receptor-binding risks for new viruses or strains, and discover linkage or motif preferences that are relevant to vaccine and therapeutic design.
- Due to its **lightweight framework and relatively small model size**, Glyde can be integrated into existing computational pipelines in virology and glycobiology. This work also creates a **foundation for reproducible glycan machine learning benchmarks and has the potential to accelerate the development of next-generation vaccines** for influenza, coronaviruses, and other viral threats. In the long term, this approach could scale to broader glycomics datasets and support more generalizable models of host–pathogen interactions.

MEDIA & REFERENCES

1) Davey, R. "What Are Glycans?" *News-Medical*, 2021.
2) Elicityl-OligoTech. "2,6-Sialylated Complex-Type N-Glycan (>90% HPLC)." *Elicityl-OligoTech Product Page*.
3) Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., and Polosukhin, I. "Attention Is All You Need." *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
4) Apple. "Apple Unveils M3, M3 Pro, and M3 Max, the Most Advanced Chips for a Personal Computer." *Apple Newsroom*, October 30, 2023.
5) Harris, M. "NVIDIA DGX-1: The Fastest Deep Learning System." *NVIDIA Technical Blog*, 2016.
6) National Center for Biotechnology Information. "SNFG Tools." *NCBI Glycans*.
7) Varki, A. "Biological Roles of Glycans." *Glycobiology*, vol. 27, no. 1, 2017, pp. 3–49.
8) Coff, L., Chan, J., Ramsland, P. A., and Guy, A. J. "Identifying Glycan Motifs Using a Novel Subtree Mining Approach." *BMC Bioinformatics*, 2020.
9) Carpenter, E. J., et al. "GlyNet: A Multi-Task Neural Network for Predicting Protein–Glycan Interactions." *Chemical Science*, 2022.
10) Lundstrøm, J., Korhonen, E., Lisacek, F., and Bojar, D. "LectinOracle: A Generalizable Deep Learning Model for Lectin–Glycan Binding Prediction." *Advanced Science*, 2022.
11) Burkholz, R., Quackenbush, J., and Bojar, D. "Using Graph Convolutional Neural Networks to Learn a Representation for Glycans." *Cell Reports*, 2021.
12) Neelamegham, S., et al. "Updates to the Symbol Nomenclature for Glycans Guidelines." *Glycobiology*, vol. 29, no. 9, 2019, pp. 620–624.
13) GlyGen Consortium. *GlyGen Glycan Array Data Repository*. GlyGen, University of Georgia.
14) Consortium for Functional Glycomics. *CFG Glycan Array Experiments*. Consortium for Functional Glycomics.
15) Fujita, Akihiro, et al. "The International Glycan Repository GlyTouCan Version 3.0." *Nucleic Acids Research*, vol. 49, no. D1, 2021, pp. D1529–D1535. https://doi.org/10.1093/nar/gkaa1089.
16) Park, D., Lee, J., et al. "Increasing the α2,6-Sialylation of Glycoproteins." *Semantic Scholar*.
17) Sigma-Aldrich. "N-Glycans." *Sigma-Aldrich Technical Article*.
18) Bohne-Lang, A., and von der Lieth, C. W. "GlyProt: In Silico Glycosylation of Proteins." *Nucleic Acids Research*, 2005.
19) Kearns, F. L., et al. "Glycan–Protein Interactions in Viral Pathogenesis." *Nature Reviews Microbiology*, 2020.